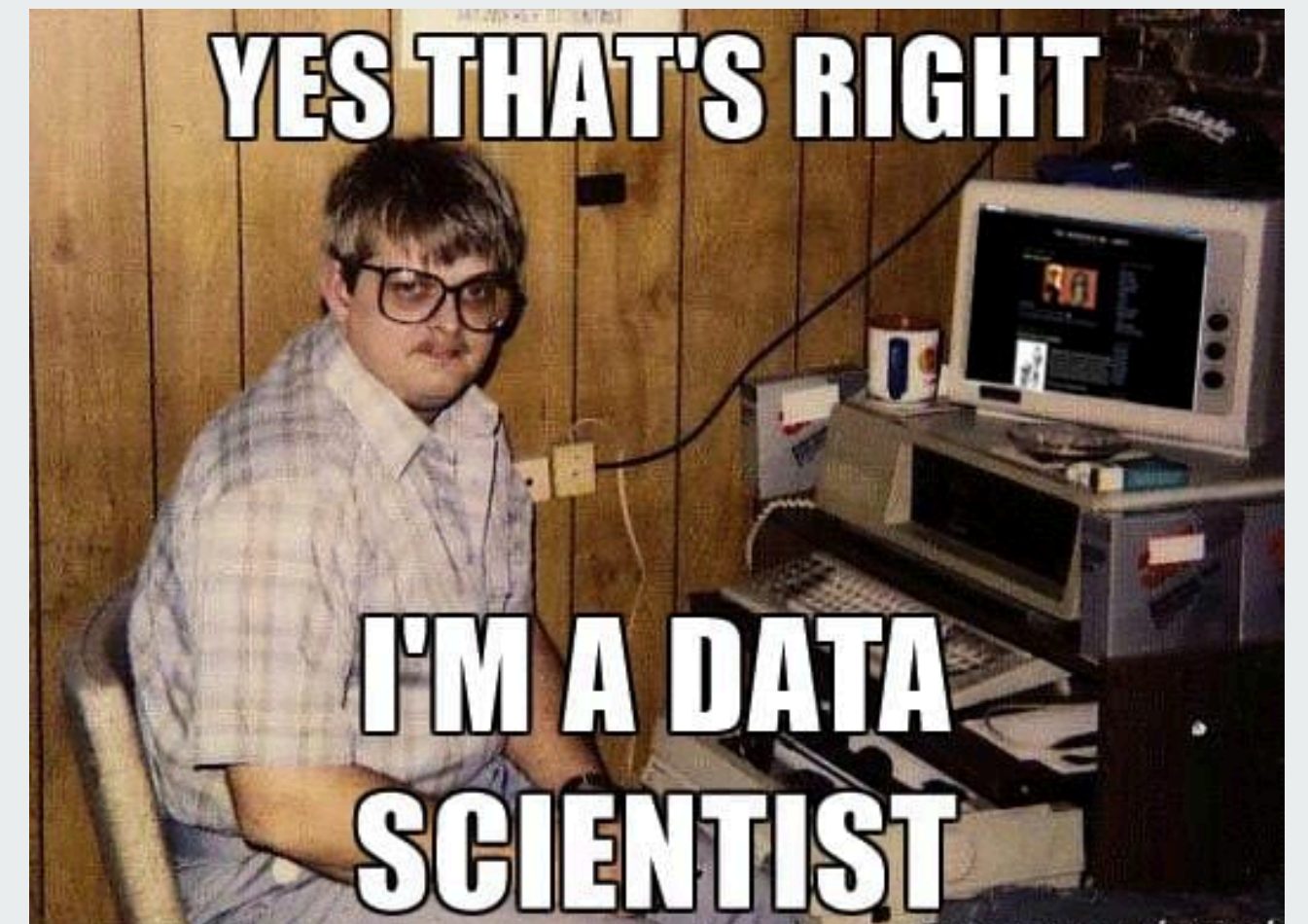




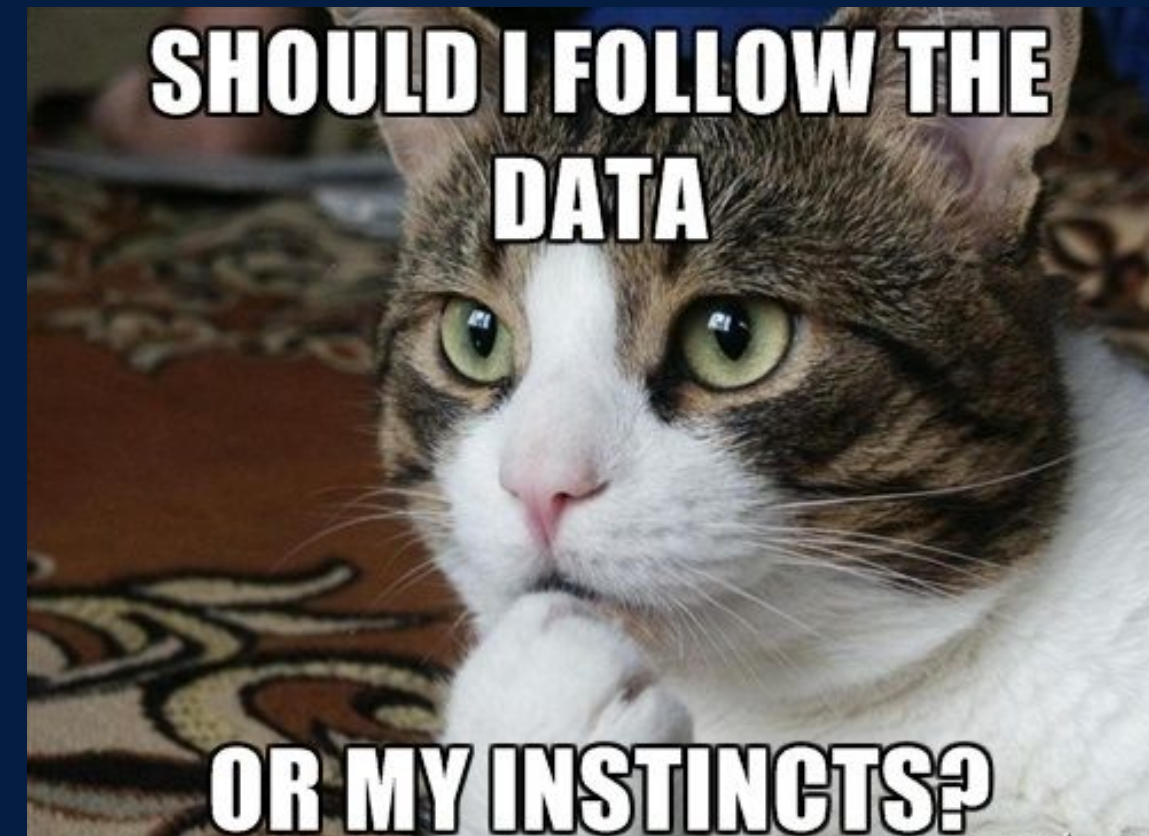
CLUB OF DATA ENGINEERS
SCHOOL OF INFORMATION SCIENCES

So, You wanna learn Data Science ?

By : Salim EL MARDI



Data ?





What is Data ?

- Data is raw information — numbers, text, images, or sounds.
- It comes from people, machines, or the web.
- Alone it means little, but once analyzed, it gives insights and knowledge.
- Two types: structured (tables, spreadsheets) and unstructured (tweets, photos).
- It's the fuel of AI and machine learning — no data, no intelligence.



Data vs Information

- Data → Raw facts and figures (e.g., “25, 30, 45”).
- Information → Processed data that has meaning (e.g., “The average age is 33”).
- Data is unorganized, while information is structured and useful.
- Think of it like this:
 -  Data = puzzle pieces
 -  Information = the completed picture

**Data science , Data Engineering, Data Analyst ,
Machine Learning Engineer, ...**





Data scientist

- Combines statistics, coding, and domain knowledge to model data.
- Builds predictive models and runs experiments.
- Often uses machine learning to find deeper insights.



Data Engineer

- Builds and maintains data pipelines and infrastructure.
- Ensures data is collected, cleaned, and stored efficiently.
- Works with databases, cloud platforms, and big data tools (e.g., Spark, Hadoop).



Data Analyst

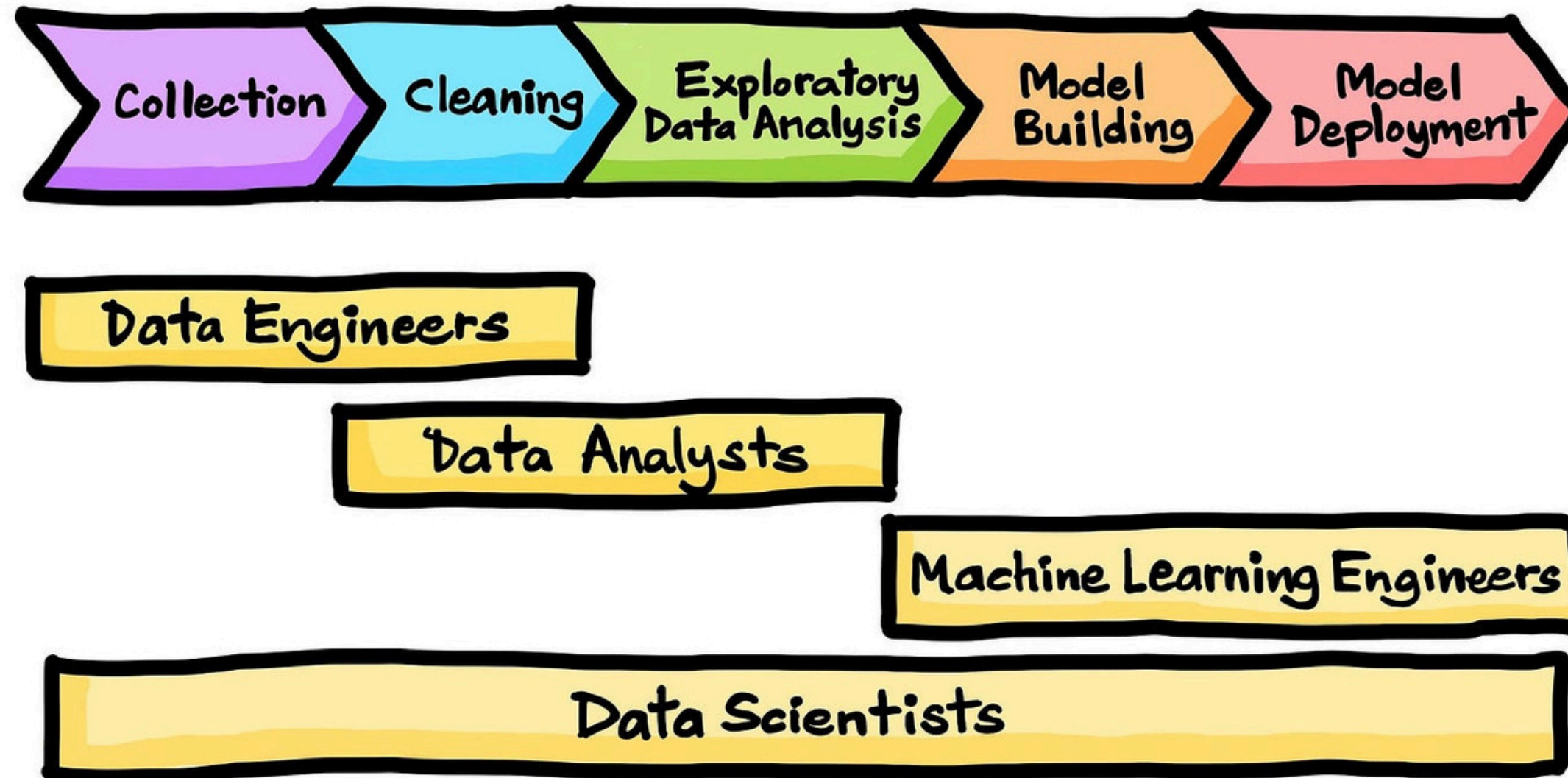
- Explores and visualizes data to find patterns and insights.
- Helps businesses make data-driven decisions.
- Uses tools like Excel, SQL, Power BI, or Python.



ML Engineer

- Turns models into real-world products or systems.
- Optimizes and deploys ML models at scale (e.g., in apps or APIs).
- Bridges software engineering and AI modeling.

Data Science Project lifecycle

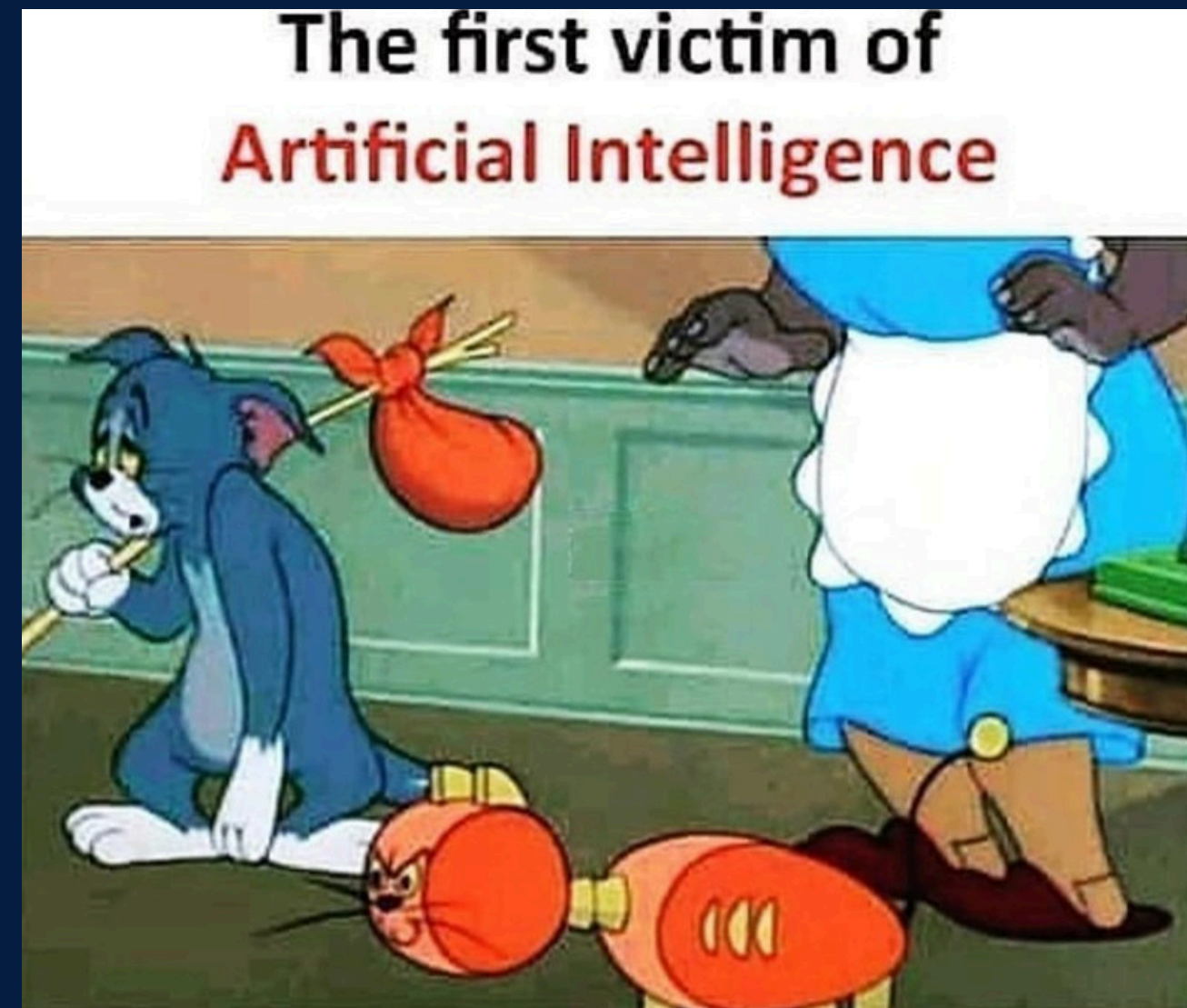




Real World Applications of Data Science

- **Healthcare** – Disease prediction, medical imaging, drug discovery.
- **Finance** – Fraud detection, algorithmic trading, credit scoring.
- **E-commerce** – Recommendation systems, customer analytics.
- **Transportation** – Self-driving cars, route optimization.

What is AI ?



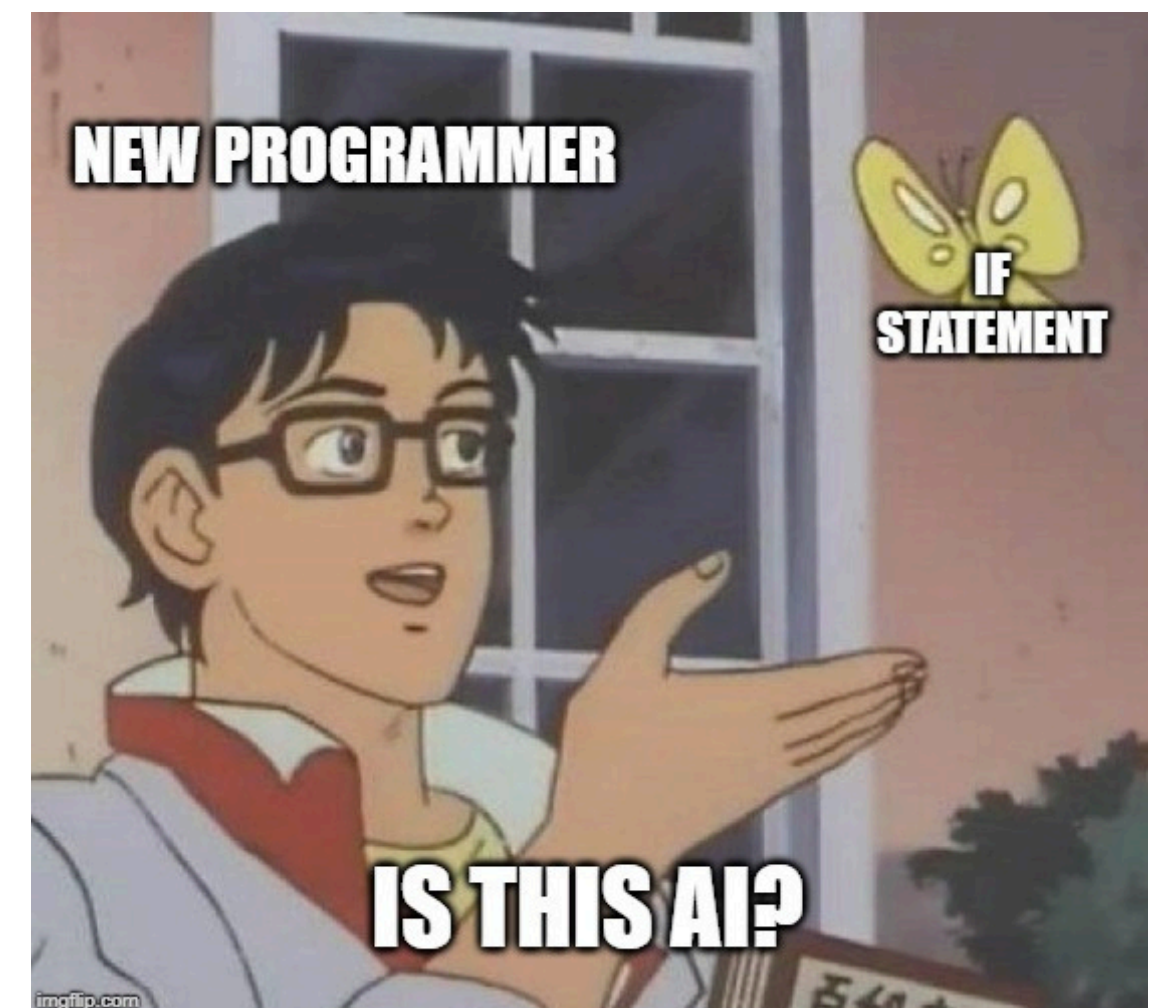


AI

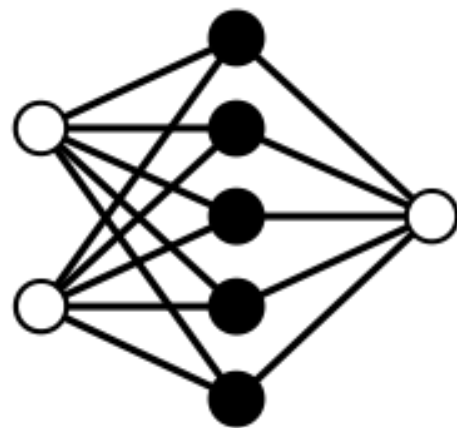
Artificial Intelligence (AI) is the science of building systems that can think, learn, and act like humans.

AI vs Algorithms

AI	Algorithms
Learns from data	Follows fixed rules
Adapts and improves over time	Always behaves the same way

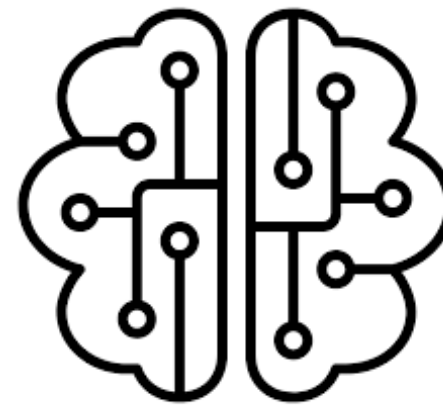


AI History Overview



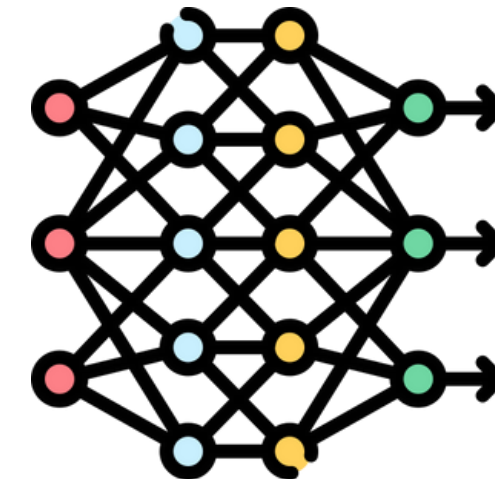
Neural Networks

1950 – 1970



Machine Learning

1980–2010



Deep Learning

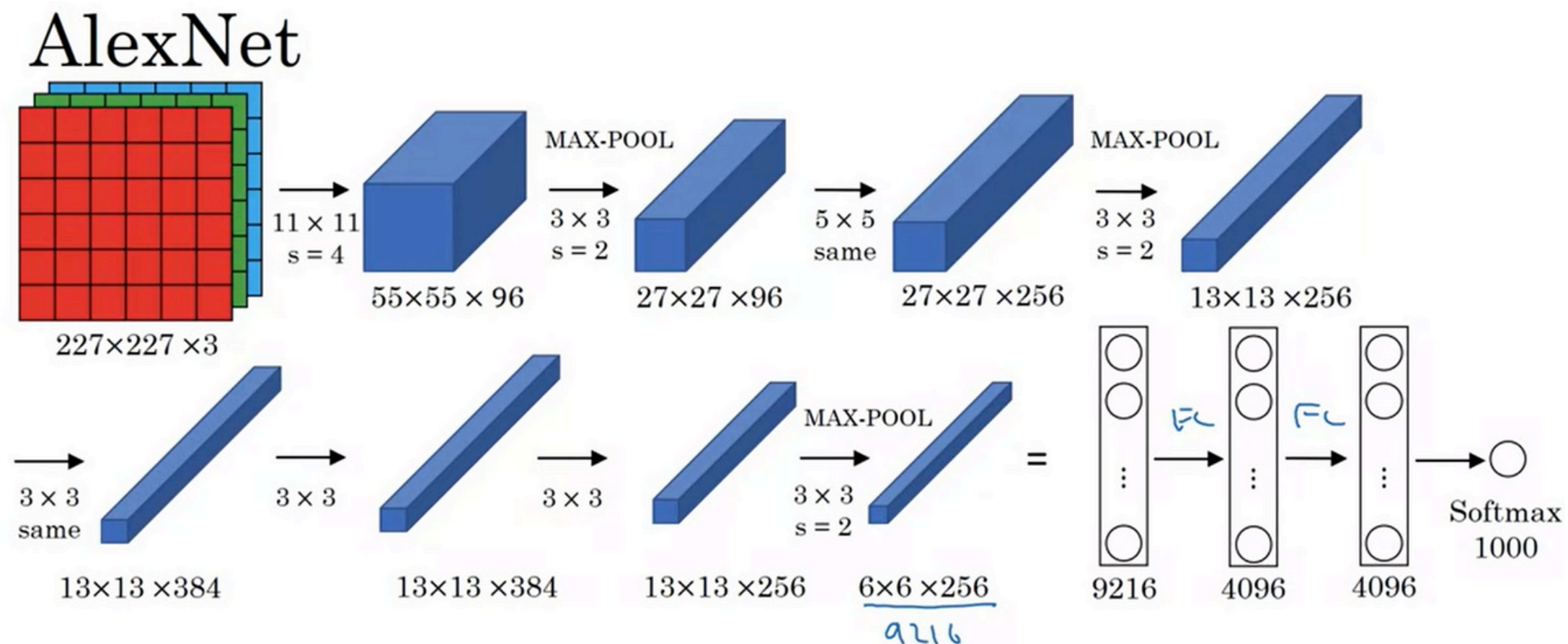
TODAY



Deep Blue beats Garry Kasparov (1977)



AlexNet revolutionizes image recognition (2012)



AlphaGo defeats Lee Sedol in Go (2016)



GPT-3 and large language models (2020)



Generative AI explosion (2022 - Now)





Resources



Online Learning Platforms

- **Coursera** – Structured courses and specializations from top universities and companies.
- **Fast.ai** – Practical, project-based courses focused on coding and deep learning.
- **DeepLearning.AI** – Industry-leading short courses and specializations in AI and ML.
- **Kaggle Learn** – Interactive micro-courses for quick, hands-on skill building.
- **Medium / Towards Data Science** – Articles, tutorials, and case studies from professionals in the field.



Skill Application & Collaboration

- Kaggle Competitions
- Collaborative Projects
- Mini Local Competitions
- External Competitions
- Hands-on Workshops by Experts

Projects

Small Assignment



Python overview (Kaggle Learn)



Learn Python Tutorials

Learn the most important language for data science.

 kaggle.com

**Thank You For
Your Attention !**